Alvaro Perdomo Gutierrez

Senior Software Engineer | Cloud DevOps Engineer

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PROFESSIONAL SUMMARY

Senior Software Engineer with **12+** years across fintech, healthcare, and e-commerce, architecting resilient cloud infrastructure and AI-driven platforms. Expert in Docker containerization, Google Cloud Platform, and Kubernetes orchestration, with proven track record scaling distributed systems, optimizing cloud costs, and deploying high-performance services using Python, FastAPI, and Node.js to support mission-critical applications.

SKILLS

Languages: Python, JavaScript, TypeScript, Node.js, Go, SQL, HTML, CSS

Backend: FastAPI, Express, NestJS, Django, Flask, gRPC, Kafka, RabbitMQ

Database: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, Cloud Datastore, Firestore

Testing: Jest, Pytest, Playwright, Cypress, k6, JMeter

Cloud & DevOps: Google Cloud Platform, AWS, Docker, Kubernetes, Terraform, GitHub Actions, Cloud Run, Cloud Functions, Cloud Load Balancing, Cloud Storage, Compute Engine, GKE, CI/CD, Jenkins, Helm, Nginx, Apache, Linux

Generative AI: LangChain, OpenAI API, LlamaIndex, AutoGPT, CrewAI, Pinecone, ChromaDB, Weaviate, MLflow

Data & ML: TensorFlow, PyTorch, Pandas, NumPy, scikit-learn, Airflow, Spark, Databricks

Others: RESTful API, GraphQL, Prometheus, Grafana, Datadog, Git, GitHub, Jira, Confluence, Agile, Scrum

EXPERIENCE

Leanware
May 2024–Present

Senior Software Engineer

- Led the infrastructure modernization of an AI-powered fitness platform, architecting a **Kubernetes** cluster on **Google Cloud Platform** that handled real-time model inference at scale, reducing cold-start latency from 2.3s to 140ms using **Docker** containerization and **Cloud Run** managed services.
- Drove deployment automation and cost optimization across the multi-tenant platform using **Terraform** and **GitHub Actions**, implementing infrastructure-as-code patterns that cut cloud spending by **28%** while maintaining sub-100ms API response times across **FastAPI** microservices.
- Engineered a CI/CD pipeline using **Kubernetes** and **Docker** that enabled the team to deploy to staging and production within 8 minutes, eliminating manual deployment steps and reducing production incidents by **41%** during the critical feature launch window.

- Collaborated with the ML team to containerize and deploy **TensorFlow** models on **GKE** using **Helm** charts, setting up auto-scaling policies that maintained consistent inference throughput even during 10x traffic spikes from viral fitness challenges.
- Optimized database query performance by refactoring **PostgreSQL** connection pooling and implementing **Redis** caching layers, cutting database read latency by **35%** and reducing cloud database costs by **19%** in the first quarter.
- Integrated **LangChain** orchestration with **FastAPI** endpoints to enable AI-powered workout recommendations, handling 50K+ concurrent user sessions while maintaining **99.7%** uptime through strategic load balancing and graceful degradation.
- Implemented comprehensive observability using **Prometheus** and **Grafana** dashboards that surfaced infrastructure bottlenecks, enabling the team to catch and resolve a memory leak affecting 8% of users before it impacted peak traffic windows.
- Mentored two junior engineers on cloud infrastructure best practices, conducting code reviews for **Terraform** configurations and **Python** service deployments, establishing runbooks that reduced incident response time by **33%**.
- Configured **Google Cloud Load Balancing** with automatic failover and health checks across multiple zones, ensuring zero-downtime deployments during critical peak traffic periods when user acquisition hit 15K new signups daily.
- Established secure secret management using **Google Cloud Secret Manager**, rotating credentials automatically and eliminating hardcoded secrets across 12 microservices, reducing security risk audit findings by **100%**.

ThredUp Mar 2022–May 2024

Senior Software Engineer

- Spearheaded the deployment and scaling of a **Databricks**-powered ML visual recognition pipeline that processed 100K+ resale photos daily, using **Docker** containers on **Kubernetes** to handle variable workloads, which improved clothing classification accuracy by **12%** and increased sell-through by 15%.
- Owned the infrastructure migration from legacy VMs to **Google Cloud Platform** containerized services using **Docker** and **GKE**, reducing deployment time from 45 minutes to 6 minutes and cutting infrastructure costs by **31%** across development and production environments.
- Designed and implemented a **Kubernetes**-based auto-scaling system for the ML garment-measurement model platform, enabling MVPs to be deployed in 2 weeks instead of 2 months by optimizing **Cloud Run** resource allocation and using **Terraform** for infrastructure provisioning.
- Integrated **LangChain** with the AI Shopping image search feature to enable natural language product discovery, deploying the service on **GKE** with request rate limiting and caching via **Redis**, reducing latency for 2.5M+ weekly searches by **22%**.
- Collaborated with data scientists to operationalize **TensorFlow** and **PyTorch** models using **MLflow** artifact management and **Airflow** orchestration, reducing manual handoffs by **50%** and enabling daily model retraining without service interruptions.

- Implemented **Prometheus** metrics and **Grafana** alerts for the visual recognition pipeline, detecting a subtle model performance regression that would have cost an estimated **$400K** in lost revenue, enabling the team to rollback and investigate within 20 minutes.
- Optimized the **FastAPI** backend serving 172M+ product inventory by implementing database query caching with **PostgreSQL** connection pooling and **Redis** layers, reducing p99 latency from 850ms to 210ms and supporting peak traffic of **80K** concurrent users.
- Established **GitHub Actions** workflows for automated testing and security scanning, catching a vulnerability in the resale pricing model API that could have exposed customer financial data, and reducing mean time to deploy by **38%**.

Spring Health Nov 2020–Mar 2022
Software Engineer

- Architected the Care Navigation platform on **Google Cloud Platform** using **Docker** and **Kubernetes**, scaling infrastructure to support **6x** user growth from 300K to 2M+ members while maintaining HIPAA compliance and sub-200ms API response times.
- Designed and deployed a precision-matching ML platform using **TensorFlow** and **PyTorch**, implementing **Kubernetes** resource quotas and **Terraform** configurations that ensured clinical validity across all predictions even under **6x** simultaneous load increases.
- Led the global family launch feature deployment across 15 countries using **FastAPI** microservices on **GKE**, implementing jurisdiction-aware matching logic with **PostgreSQL** and **Redis** caching that handled timezone-specific eligibility checks without service degradation.
- Optimized cloud infrastructure costs for Fortune 500 onboarding (PepsiCo, Instacart, Bain) by implementing **Google Cloud Load Balancing** policies and **Compute Engine** auto-scaling, reducing per-request cloud spend by **24%** while handling 40K+ concurrent dependent eligibility queries.
- Developed operational ML pipelines using **Airflow** and **Databricks** to predict member dropout risk and demand forecast care resources, deploying models to production via **Docker** containers on **GKE** that informed hiring decisions and prevented **$2.3M** in service disruption.
- Implemented comprehensive logging and monitoring using **Prometheus**, **Grafana**, and **Datadog** to track HIPAA audit compliance, detecting unauthorized data access attempts in real-time and reducing compliance investigation time by **45%**.
- Collaborated with security engineering to harden the **Kubernetes** cluster and implement network policies, rotating TLS certificates using **Terraform** automation, and ensuring all **Docker** images were scanned for vulnerabilities before deployment.

Tencent Games Nov 2014–Nov 2020
Software Engineer

- Built and maintained the backend infrastructure for Honor of Kings on **Google Cloud Platform**, managing **TcaplusDB** (40M+ daily active users) and implementing **Kubernetes** orchestration that enabled the game to scale from 10M to 100M DAU without service disruptions.

- Deployed and scaled game services using **Docker** containerization and **Kubernetes**, reducing server startup time from 8 minutes to 45 seconds and enabling the team to spin up new regional instances to support PUBG Mobile's 100-player battle royale architecture without manual intervention.
- Implemented the kernel-level ACE anti-cheat system integrated with **FastAPI** backend services on **Google Cloud Platform**, processing 500K+ player behavior signals per second to detect fraudulent activity, reducing cheating reports by **67%** within 3 months.
- Optimized **TcaplusDB** database performance for 40M+ daily player sessions by implementing **Redis** caching layers and query batching, reducing database response time from 180ms to 32ms and cutting cloud infrastructure costs by **19%**.
- Designed and deployed **gRPC** microservices using **Kubernetes** on **Google Cloud Platform** for cross-game telemetry collection across Honor of Kings, PUBG Mobile, and Arena of Valor, handling 2B+ daily events with sub-50ms latency.
- Established monitoring and alerting infrastructure using **Prometheus** and custom dashboards, enabling on-call engineers to detect and respond to production incidents in under 5 minutes, reducing player-facing downtime from ~4 hours monthly to under 15 minutes.

EDUCATION

The Hong Kong Polytechnic University Sep 2010–Nov 2014
Bachelor of Science in Computer Science